Daily Drilling Report
SPD19B-03
3/17/2014

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
7.0	24"	13.00	72.83	31.70	92.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
26" CP, 178.00mKB	0.00	19,405.90	606,913.73		5.60
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
579.00	722.00		143.00	19.41	7.4

Operations at Report Time
Drilling @ 735m.

Operations Summary
Drilling 24" hole f/ 579m to 722m

Operations next 24hrs
Cont' drilling 24" hole to TD. Sweep hole.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Mohammad Ali Rezaei	Senior Drilling Sup.
Jabbar Mohammadi	Night Drilling Sup.
Hamed Daneshkhah	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	3/9/2014	3/31/2014
BOP Test		
Chemical Spill Drill		
Fire Drill	3/9/2014	3/11/2014
H2S Drill	3/16/2014	3/31/2014
Heli Crash Drill	2/20/2014	3/14/2014
Medical Drill		
MOB Drill	3/1/2014	3/23/2014
Pit Drill		
Trip Drill		

STOP Cards

Number of Reports: 41

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment

00:00	00:00	24....	24"	D	03	P	Drilling 24" hole f/ 579m to 722m w/ 1100-950 gpm / 1650-2100 psi / 35-40 kips/ 35-70 rpm & 7 - 23 Kft-lbs.
							
							>>> Sweeping Sequence f/ 571m:
							** At the Middle of each joint: Pump 50 bbl GG.
							** At the End of each joint: Pump 30 bbl GG + Circulate 3 min + 50 bbl PHG.
							** At the End of Stand: Pump 50 bbl GG + Circulate 5 min + 50 bbl PHG. B/R & R/D the stand.
							** Take survey one single off bottom.
							** Spot 40 bbl PHG + 30 bbl GG (5 bbl short displacement). Make connection. After connection, first rotate the string then bring up flow rate.
							
							>> Drill ahead w/ complete loss, fill up annulus w/ SW.
							
							>>> Sweeping Sequence f/ 670m (Expected Top of Sachun Formation):
							** At the Middle of each joint: Pump 40 bbl GG.
							** At the End of each joint: Pump 40 bbl PHG.
							** At the End of Stand: Pump 40 bbl GG + Circulate 5 min + 40 bbl PHG. B/R & R/D the stand.
							** Take survey one single off bottom.
							** Spot 30 bbl PHG + 20 bbl GG.
							
							Mar 18, 2014
							00:00 - 06:00 Cont' drilling 24" hole f/ 722m to 735m w/ 950 gpm / 1680 psi / 40 kips/ 55 rpm & 6 - 18 Kft-lbs.

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Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
722.00	Sea Water	8.60				
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
1.50	5.	353	20.0	1,017.0	22.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
EP-HBJ	Chopper	3/17/2014 16:09	3/17/2014 16:13	Pax in: 0 & Pax out: 1
DP3	Supply Boat	2/23/2014 23:30		Standby Anchorage.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Barite Bulk	16.3	221.0	204.7
Bentonite Bulk	5.5	105.0	71.0
Bentonite BB	8.0	45.0	37.0
Bio-Cide	12.0	139.0	77.0
Citric Acid	6.0	210.0	74.0
Caustic Soda	12.0	427.0	249.0
Soda Ash	6.0	151.0	45.0
Lime	9.0	150.0	57.0
Guar Gam MP	83.0	1,421.0	1,034.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Diesel Fuel	9.0	0.0	181.0
Potable Water	21.0	24.0	90.0
Drill Water	97.0	0.0	248.0
Cement G	0.0	0.0	183.0
Bentonite Bulk	5.0	0.0	63.0
Bentonite BB	8.0	0.0	37.0
Barite Bulk	17.0	0.0	192.0
Barite BB	0.0	0.0	0.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
3	24	Varel	ETR14J	1250452	InserT Bit	435	24/24/24/20
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
1.60	28.86	99,813.00	491.00	722.00	231.00	8.0	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
850	1,100	1,250.0	2,100.0	22	40	40	80
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
40.36	28.86	100.00	0.00	262	240	219	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
579.00	633.00	3.35	23,000.0	1,100	50	2,100.0	35	230
633.00	690.00	5.46	17,000.0	1,100	40	2,100.0	35	240
690.00	719.00	8.91	17,000.0	1,000	70	1,780.0	40	240
719.00	722.00	1.69	16,000.0	950	55	1,650.0	40	240

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
13 Jts 5 7/8" HWDP	5 7/8	4.250	120.97	256.31
X/O	7	2.810	0.79	135.34
2 Jts 8 1/4" DC	8 1/4	2.875	18.75	134.55
8" Cougar DA Jar	8	3.500	5.84	115.80
5 Jts 8 1/4" DC	8 1/4	2.875	47.24	109.96
X/O	9 5/8	3.000	1.22	62.72
3 Jts 9 1/2" DC	9 1/2	3.000	28.29	61.50
17 1/2" Stabilizer	17 1/2	3.000	2.47	33.21
9 1/2" NMDC	9 1/2	4.000	9.45	30.74
9 1/2" TM Sub (MWD & Gamma probes inside)	9 1/2	3.125	2.15	21.29
9 1/2" NMDC	9 1/2	4.000	9.45	19.14
9 1/2" Restrictor sub (32/32" restrictor)	9 1/2	4.250	0.91	9.69
23" Stabilizer	23	3.000	2.86	8.78
9 1/2" Float Sub (Float valve inside)	9 1/2		0.79	5.92
9 1/8" Shock Sub	10 1/8	2.625	4.51	5.13
24" Insert Bit	24		0.62	0.62

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Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)

Applied Surface Pressure (psi)	Volume Pumped (bbl)	Current MAASP	Kick Tolerance	Dens Fluid (lb/gal)	Leak Off Pressure (psi)

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
579.73	240.70	18.30	570.07	-51.32	-61.58	79.94	1.48	0.20
608.69	242.20	18.00	597.59	-55.63	-69.50	88.90	0.58	-0.31
637.53	243.40	17.90	625.03	-59.70	-77.40	97.69	0.40	-0.10
666.40	245.00	19.20	652.40	-63.69	-85.67	106.74	1.45	1.35
695.34	246.40	19.30	679.72	-67.61	-94.37	116.09	0.49	0.10

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
Asmari		374.50	372.44
Jahrum		493.00	487.29

General Remarks

Comment

>>> Requirements: (23" Stabilizer) + (Shock Sub) + (300 MT DW) + (PDF MR#6).

>>> Sent:Nil.

>>> Received: Nil.